Lesson plan · for the teacher

Tricking AI and defending it: unit check

AI security · Level 2 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Orders hidden in the data
- Jailbreaks: talking a chatbot round
- Guardrails: rules around the model
- Ask before it acts

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which AI tool at your school reads text other people wrote?
- Which question surprised you, and why?
- What will you ask the next time someone offers the school an AI agent?

Exit ticket

- One defence I will ask for, and where I will ask for it

Try it in class tomorrow

Pick the defence from this unit your school needs most (outside text as data, reporting what slips through, a check on the action, or only the tools a job needs) and ask about it the next time anyone suggests a new AI tool.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **What is prompt injection?**

- ◯ An order hidden in text an AI reads, which it may follow
- ◯ A virus injected into a phone
- ◯ A typing mistake in a prompt

2 **A language model can always tell your instructions apart from words inside a document it is reading.**

- ◯ True
- ◯ False

3 **Match each term to its meaning.**

Direct injection	• • • •	Testing a chatbot on purpose, with permission
Indirect injection	• • • •	Talking a chatbot round its safety rules
Jailbreak	• • • •	An order hidden in an email the AI summarises
Red teaming	• • • •	An order typed into the chat

4 **Why can a reworded request get past a chatbot's refusal?**

- ◯ Its refusals are learned from the wordings it saw
- ◯ The chatbot gets tired after a few questions
- ◯ Rewording deletes its safety rules

5 **Pupils made the school chatbot write something unkind. Which responses make sense? Choose all that apply.**

- ☐ Record what was tried and what got through
- ☐ Report it to the provider
- ☐ Add a check or rule, then test again
- ☐ Hold a contest to find more